

Simulating quantum spin ice in a small box: the winding-free campaign, explained from the beginning

Pedagogical companion to `twist_qsi_demo`

July 2026

Contents

1	Why this document exists	2
2	The stage: what quantum spin ice is	3
2.1	Spins on a pyrochlore lattice	3
2.2	Three characters emerge	3
2.3	The Hamiltonian, for reference	4
3	The instrument: exact diagonalization on a torus	4
3.1	Why small boxes	4
3.2	Two kinds of loops on a doughnut	4
4	The villain: the box invents a process stronger than the physics	5
4.1	Counting steps	5
4.2	The measured symptom	6
5	First attempt: twist averaging — and why it fails	6
5.1	The idea	6
5.2	The failure, precisely	7
6	The fix: a label that sees topology, and the mask	8
6.1	Transport: the bookkeeping number	8
6.2	The mask	8
6.3	What the mask never asks: configurations, not eigenstates	9
6.4	Why this is safe, and why nothing finer exists	9
6.5	Three theorems, in plain words	9
7	A toolkit interlude: folding the universe into a small matrix	10
8	Strong coupling: the band dissolves, and the map survives	11
8.1	What breaks, and when	11
8.2	What survives	12
8.3	Life inside the continuum: tracking without repairing	12

9	One honest Hamiltonian: the counterterm, its limits, the splice	14
9.1	Why there is nothing to cut: the ice block of H is empty	14
9.2	The construction	14
9.3	The obstruction theorem, and honesty about the residue	15
9.4	What the counterterm cannot do	15
9.5	A counterterm-free future: folding from both sides	16
9.6	The third peak: three bumps, one name	19
9.7	What the crossing means — and what it does not	20
10	Turning the second dial: what $J_{\pm\pm}$ actually does	22
10.1	What a pair flip costs	22
10.2	Three measured effects, and a squeeze	23
10.3	Why it stops: the dial destroys what it moves	23
10.4	The verdict, and one honest complication	24
11	What the campaign established	24
12	The cluster is the error bar	25
12.1	First steps onto FCC-32: calibrate, then scale	26
13	The road ahead: beating finite size at strong coupling	27
13.1	Residual 1: topology — already solved	27
13.2	Residual 2: matched winding pairs — solved by geometry	27
13.3	Residual 3: the coarse spinon continuum — solved by twists, reborn	27
13.4	Honesty items	28
13.5	The validation ladder	29
14	Where the method ends: the Heisenberg point, and beyond	29
14.1	The Heisenberg point	29
14.2	Beyond, in the XXZ model: the mask retires	30
14.3	Beyond, in the XYZ model: the mask reborn	30
15	Wrong turns, kept on purpose	31
16	Cast of characters	34

1 Why this document exists

This is the story of a long computational campaign, told so that a reader with *no* mathematical background can follow every step. The campaign asks a deceptively simple question:

Can we trust a computer simulation of a quantum magnet when the simulation is forced to live in a tiny box?

For the material family we care about — *quantum spin ice*, realized in cerium pyrochlore magnets such as $\text{Ce}_2\text{Zr}_2\text{O}_7$ and $\text{Ce}_2\text{Hf}_2\text{O}_7$ — the answer turned out to be “**no, not naively**”: the tiny box invents a fake physical process that is *stronger* than the real one, and every naive simulation measures the fake. The campaign (i) identified the fake process precisely, (ii) proved that the popular fix does *not* work, (iii) constructed a fix that provably works, (iv) pushed that fix into the strongly interacting regime where the usual approximations collapse, and (v) mapped out what is left to do. Along the way it collected several wrong turns that are as instructive as the successes; they are all documented here.

Every number quoted in this document was computed and cross-checked in the campaign; nothing is schematic except the drawings.

2 The stage: what quantum spin ice is

2.1 Spins on a pyrochlore lattice

The magnetic atoms in these materials sit on a *pyrochlore lattice*: a network of corner-sharing *tetrahedra* (a tetrahedron is a pyramid with four triangular faces and four corners). Each corner hosts one magnetic moment — a “spin” — which you can picture as a tiny arrow that is only allowed to point either *into* the tetrahedron or *out of* it.

The dominant interaction, called J_{zz} , is a simple rule of thumb:

each tetrahedron is happiest when exactly two of its four arrows point in and two point out.

This is the famous **ice rule**, named for its exact analogy with the positions of hydrogen atoms in ordinary water ice.

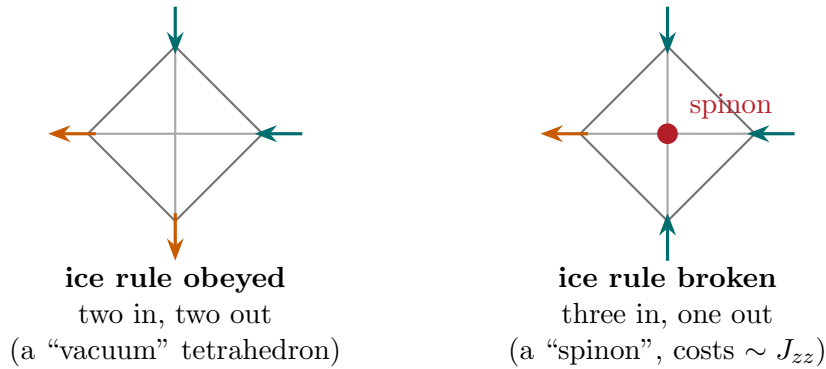


Figure 1: One tetrahedron of the pyrochlore lattice, seen from above. Each of the four corners carries a spin that points in (teal) or out (orange). Left: an ice-rule configuration. Right: flipping one spin creates a defect — a *spinon* — with an energy cost of order J_{zz} .

The ice rule does not pick a single winner. On a lattice of N spins there is an enormous number of different arrangements that all satisfy “two in, two out” on every tetrahedron — for our 16-site simulation cell there are exactly 90 of them, for the 32-site cell 2970. This collection of equally happy states is called the **ice manifold**. All the interesting low-temperature physics happens *inside* this collection.

2.2 Three characters emerge

Quantum mechanics lets the system tunnel between different ice configurations, and out of this tunneling emerges a spectacular fiction: the system behaves, at low energy, like a miniature universe with its own electromagnetism. Three characters matter for us:

Spinons. Break the ice rule on one tetrahedron (three-in-one-out) and you create a particle-like defect. Spinons are the “electric charges” of the emergent electromagnetism. They are expensive (energy $\sim J_{zz}$) and show up in measurements as a broad feature at temperatures $T \sim J_{zz}/4$.

The photon / ring exchange. Within the ice manifold, the cheapest quantum motion is a *ring exchange*: six spins arranged around a hexagonal loop flip together (in $\rightarrow$ out $\rightarrow$ in $\rightarrow$...

becomes out→in→out...). This collective six-spin flip is the matrix out of which the emergent “light” (the photon) is built. Its strength is tiny:

$$g_6 = \frac{12|J_{\pm}|^3}{J_{zz}^2}, \quad (1)$$

where $J_{\pm}$ is the quantum (spin-flipping) part of the interaction. The power *three* appears because the loop must be built in three elementary steps (more on this below).

Flux. Going around a hexagon, the quantum amplitudes can add up as if a magnetic flux threads the loop. For $J_{\pm} < 0$ (the sign relevant to the cerium materials) theory predicts π -*flux*: each hexagon behaves as if half a flux quantum pierces it. Detecting this flux is one of the big experimental goals.

In plain words

A magnet whose spins obey an ice rule hides a fictional universe: defects act like electric charges (spinons), collective six-spin flips act like light (photons), and hexagons can carry magnetic flux. The photon scale g_6 is minuscule — it involves $|J_{\pm}|^3$, the cube of an already small number — so any pollution of the simulation at a *lower* power of $J_{\pm}$ will drown it. Keep that sentence in mind; it is the whole story.

2.3 The Hamiltonian, for reference

For completeness (skip freely), the model is the XXZ pyrochlore Hamiltonian

$$H = J_{zz} \sum_{\langle ij \rangle} S_i^z S_j^z - J_{\pm} \sum_{\langle ij \rangle} (S_i^+ S_j^- + S_i^- S_j^+), \quad (2)$$

with $J_{zz} > 0$ fixed to 1 (it just sets the unit of energy) and the single dial $J_{\pm}$, typically a few percent of J_{zz} in real materials, negative for the cerium pyrochlores. S^z is the in/out arrow; $S^{\pm}$ flips it.

3 The instrument: exact diagonalization on a torus

3.1 Why small boxes

“Exact diagonalization” (ED) means: write down *every* possible spin configuration, build the full table of quantum amplitudes between them, and let the computer find the exact energy levels. No approximation whatsoever — that is its glory. Its curse is size: with N spins there are 2^N configurations. Sixteen spins give 65 536 states (easy); 32 spins give 4.3 billion (heroic); 48 spins is beyond any computer for the full problem. So ED lives in small boxes, and to avoid fake surfaces we wrap the box around: the right edge is glued to the left edge, the top to the bottom, front to back. A box glued this way is a **torus** — a doughnut (in three dimensions, a triple doughnut).

3.2 Two kinds of loops on a doughnut

On a doughnut there are two fundamentally different kinds of closed loops, and this distinction is the geometric heart of the entire campaign.

Contractible loops can be shrunk continuously to a point without ever leaving the surface. These loops exist in the infinite crystal too. The hexagonal ring exchange — the photon — lives on contractible loops.

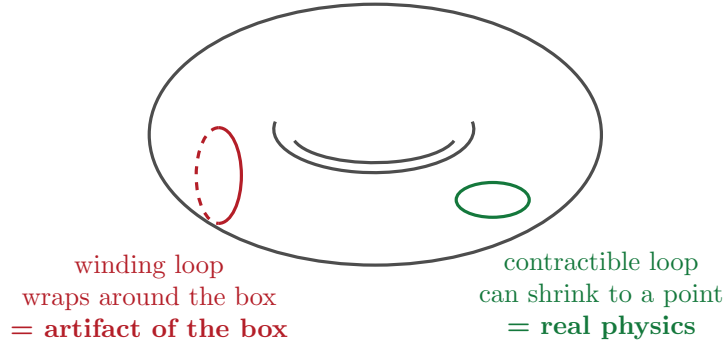


Figure 2: On a torus (a box with glued edges) a loop either shrinks to a point (green) or wraps around the box and cannot shrink (red). In an infinite crystal only the first kind exists. Everything the box invents lives on the second kind.

Winding loops wrap around the doughnut. They exist *only because the box has been glued shut*. In the infinite crystal there is nothing they correspond to. Any process that lives on a winding loop is, by construction, an artifact of the simulation geometry.

The number of times a loop wraps around each of the three glued directions is called its **winding** $w = (w_x, w_y, w_z)$, three whole numbers. Contractible loops have $w = (0, 0, 0)$. A deep mathematical fact (we will meet it again in Section 6) says this triple of integers is the *only* topological label a loop on a torus carries — there is no finer distinction.

4 The villain: the box invents a process stronger than the physics

4.1 Counting steps

How strong is a collective flip process? In quantum mechanics, a process that requires k elementary spin flips, each of strength $J_{\pm}$, passing through intermediate states that cost energy $\sim J_{zz}$, has strength roughly

$$\frac{(\text{flip strength})^k}{(\text{energy barrier})^{k-1}} = \frac{|J_{\pm}|^k}{J_{zz}^{k-1}}.$$

Each flip is a “step up the ladder”, and cheaper processes need fewer steps. The hexagon needs six spins flipped, done in three neutral steps (each step flips a $+/-$ pair), giving the $|J_{\pm}|^3$ of Eq. (1).

Here is the catastrophe. On the small torus there exist closed loops of only *four* sites that respect the ice rule — but every single one of them **wraps around the box**. (In the infinite pyrochlore lattice the shortest ice-respecting loop is the hexagon; the four-site loop only closes because the glued boundary gives it a shortcut.) A four-site loop needs just two neutral steps:

$$g_4 = \frac{4J_{\pm}^2}{J_{zz}}. \quad (3)$$

The ratio of fake to real is

$$\frac{g_4}{g_6} = \frac{J_{zz}}{3|J_{\pm}|}, \quad (4)$$

which *grows* as the coupling gets smaller — precisely in the physically relevant regime. At $J_{\pm} = -0.05$ the artifact is about seven times stronger than the photon; at $J_{\pm} = -0.01$, thirty-three times. And the box is full of these loops: the 16-site cell hosts 36 winding four-site loops and 48 boundary-wrapping hexagons, against only 16 genuine contractible hexagons.

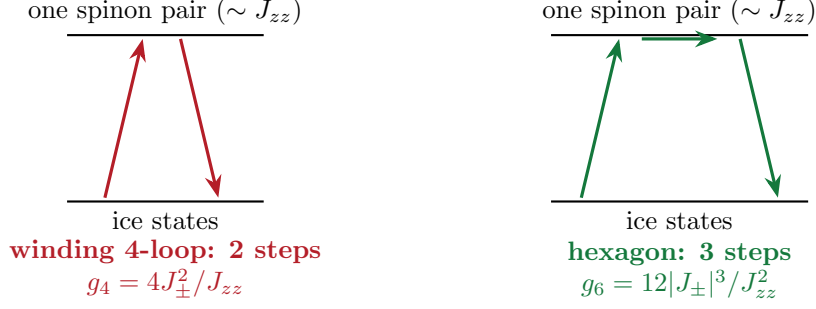


Figure 3: Why the artifact beats the physics. The fake wrap-around loop completes in two steps, the genuine hexagon needs three. Every extra step costs one more factor of the small number $|J_\pm|/J_{zz}$.

4.2 The measured symptom

The clearest thermodynamic fingerprint of the photon sector is a bump (a “Schottky anomaly”) in the specific heat $C(T)$ at a temperature set by the ring-exchange scale. The campaign measured, across the whole coupling range, where naive ED puts this bump:

	peak follows	measured exponent	ideal exponent
naive periodic ED	g_4 ($T_{\text{peak}}/g_4 \approx 1.9 \rightarrow 1.5$)	$ J_\pm ^{1.99}$	2
winding-free ED	g_6 ($T_{\text{peak}}/g_6 \rightarrow 1.02$ as $J_\pm \rightarrow 0$)	$ J_\pm ^{3.10}$	3

Measured against the photon scale g_6 , the naive peak position wanders by a factor of *forty* across the scan. Naive ED does not measure the ring scale badly — it does not measure it *at all*; it measures the box.

In plain words

The simulation box invents a two-step shortcut process that only exists because the edges are glued together. The real photon is a three-step process. Since every step costs a factor of a small number, the fake two-step process dominates everything the simulation says about the photon: the low-temperature specific-heat bump, the flux, the dynamics. The fake even gets the *sign of the flux wrong*: the naive ground state shows zero-flux-like behaviour ($\langle \text{hexagon} \rangle = +0.116$) for both signs of $J_\pm$, where the true π -flux answer is negative.

5 First attempt: twist averaging — and why it fails

5.1 The idea

There is a classical trick for taming box effects, borrowed from electronic structure theory: *twisted boundary conditions*. When a spin flip crosses the glued edge, multiply its amplitude by a phase factor $e^{i\theta}$ — as if a magnetic flux θ were threaded through the hole of the doughnut. A process that wraps the box w times picks up the phase $w\theta$; a contractible process picks up nothing. So, the hope goes: **average over twists** and the wrapping processes, whose phases rotate around the clock, cancel to zero, while the physics is untouched.

The campaign’s original claim (June 2026) was exactly this. **It is wrong**, and understanding *why* it is wrong was the first real discovery.

5.2 The failure, precisely

A ring-exchange process is not one path — it is the *sum of two* distinct virtual routes (which pair of spins flips first). The twist phase attaches to each *route*, not to the loop as a whole:

$$c(\theta) = -\frac{g_4}{2} \left(e^{-iN_A \cdot \theta} + e^{-iN_B \cdot \theta} \right), \quad N_A + N_B = w.$$

Only the *sum* of the two route-phases equals the winding. The magnitude works out to $|c(\theta)| = g_4 |\cos(w \cdot \theta/2)|$ — and here is the trap: on the 16-site cell, *every* winding four-loop has one route that never crosses the glued edge at all ($N_A = 0$, phase 1). Half of the fake amplitude is twist-blind.

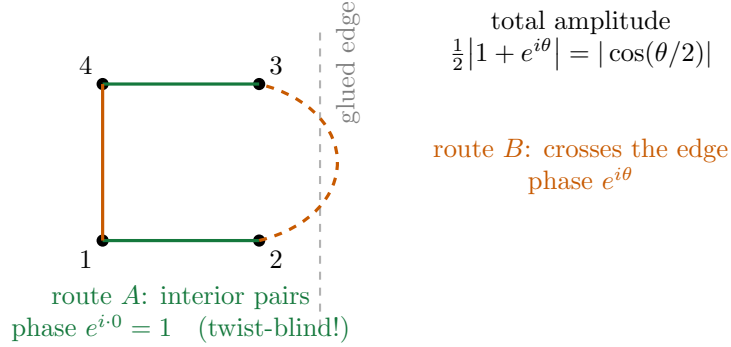


Figure 4: Why twist averaging cannot kill the wrap-around loop. The four-spin ring exchange is the sum of two virtual routes. Route *B* crosses the glued edge and feels the twist; route *A* uses only interior bonds and never sees it. No amount of averaging removes the twist-blind half.

The campaign then quantified every version of the idea:

- **Averaging measured quantities** (compute $C(T)$ at each twist, then average): weakest of all. A measured quantity depends on the *square* of amplitudes, and squares of cosines never average to zero.
- **Averaging over the special corner twists** $\theta \in \{0, \pi\}^3$: keeps exactly half of every fake coupling. Worse, a beautiful theorem (Theorem A below) shows these eight twists give *identical* spectra — they are the same physics in eight disguises, so averaging them at the level of observables does strictly nothing.
- **The best possible single twist**, $\theta = (\pi/2, \pi/2, \pi/2)$: kills 24 of the 36 fake loops exactly, but the other 12 survive at full strength. The surviving fraction of any twist-averaging scheme was proven to have a hard floor of $1/3$ — and because the surviving weight is *linear* in the choice of averaging weights, the optimum is a single point: no clever ensemble beats the best single twist.
- **The scaling test** (the damning one): with any observable-level twist scheme, the specific-heat peak still scales as $|J_{\pm}|^{1.8}$ — the artifact power — not the photon power 3. Twisting lowers the contamination; it never changes what is being measured.

Lesson learned (the hard way)

The original “theorem” that twist averaging removes winding contamination was retracted and the paper rewritten around the corrected theory. Two sub-lessons: (i) phases attach to virtual *routes*, not to the loop’s topology — always compute the object you claim vanishes;

(ii) the special twist (π, π, π) , which naively flips the sign of every wrapping bond, was verified to leave the spectrum bit-for-bit unchanged — it is a *gauge transformation*, a pure relabeling. A sign flip of a matrix element is not interference.

6 The fix: a label that sees topology, and the mask

6.1 Transport: the bookkeeping number

If phases cannot kill the winding loops, we must find something that *measures* winding exactly, and delete by hand. The campaign found it, and it is beautifully simple.

Every flip process moves magnetization around. Take the set of spins the process flips; add up their positions with a $+$ sign for raised spins and $-$ for lowered ones. This gives a vector ρ , the **transport** (or polarization displacement) of the process. The campaign proved, by explicit enumeration over every loop of both simulation cells, the exact identity

$$\rho = \frac{1}{2} w \cdot L, \quad (5)$$

where w is the winding and L the box dimensions. Read it aloud: *transport is half the winding*. A contractible loop ($w = 0$) transports nothing — the raised and lowered spins balance perfectly. A winding loop *cannot* balance: it transports exactly half a box length. There is no overlap and no ambiguity: measuring ρ is measuring topology.

6.2 The mask

The recipe is now almost embarrassingly direct. Group the 90 ice states into families (“sectors”) that share the same value of the polarization; on the 16-site cell there are 25 such sectors. Then, in the table of quantum amplitudes between ice states:

delete every entry that connects two different sectors.

That is the whole method. It is called the **mask**. By Eq. (5), an entry connecting different sectors is precisely a process with $w \neq 0$: a wrap-around artifact. An entry within a sector is a contractible process: untouched, to the last digit.

sector 1	sector 2	3	4
	kept	deleted	
	deleted		

the amplitude table between ice states, sorted by transport sector

Figure 5: The mask. Sort ice states by their polarization sector; keep the diagonal blocks (contractible physics, green), delete everything off the blocks (wrap-around artifacts, red). Deleting can only remove processes — it can never invent new ones.

6.3 What the mask never asks: configurations, not eigenstates

A question that sounds central but that the mask never answers — because it never asks it — is: *which eigenstates of H are the ice band?*

Ice-ness, in this method, is a property of *configurations*, not of energy levels. A basis pattern either satisfies two-in–two-out on every tetrahedron or it does not — a yes/no check made pattern by pattern, tetrahedron by tetrahedron, involving no diagonalization, no energy threshold, and no coupling constant. The same 90 patterns qualify at $J_{\pm} = -0.03$ and at $J_{\pm} = -0.5$. The sectors and the mask inherit this: they are combinatorial objects, fixed before any physics is computed.

“Band” identity is then an *output label, never an input classification*. A winding-free band level is, by definition, a self-consistent root of an ice-sector block (the machinery of Secs. 7 and 8), and it arrives stamped with its sector because that is where it emerged. Even when such a root sits embedded inside the continuum (Sec. 8.3), there is no ambiguity about whether it is “band”: it is a root of the ice-block map, full stop.

The contrast to draw is with the *frames* of Sec. 8 — constructions that answered the question the other way around, by hunting for the 90 most ice-like *eigenstates*. All of them die at strong coupling, and necessarily so: once bands mix, eigenstate identity is genuinely undefined. The mask survives because it staked its definition on something that cannot mix: the spin patterns themselves. (At strong coupling the physical states behind the roots are only $\sim 25\%$ ice by weight — the map does not care; the ice patterns say where a state is *anchored*, and the fold of Sec. 7 supplies everything else about it, exactly.)

6.4 Why this is safe, and why nothing finer exists

Three properties make the mask trustworthy:

1. **It only removes.** A projection deletes entries; it cannot create a process that was not there. So it cannot introduce new physics.
2. **It provably keeps the physics.** Contractible loops have $\rho = 0$ exactly (verified on every hexagon of both cells) and their amplitudes are untouched to machine precision.
3. **It is maximal.** The torus’s mathematics (its “fundamental group” is $\mathbb{Z}^3$: loops are classified by three whole numbers, full stop) guarantees that winding is the *only* topological label. A process that winds $+1$ and then -1 (net zero) is topologically identical to a contractible one and is *correctly kept* — its counterpart exists in the infinite crystal too, as an ordinary, benign finite-size correction. The mask removes everything a topological label can remove, and nothing more.

6.5 Three theorems, in plain words

Later, the twist story and the mask story were united by three exact theorems (each verified to fifteen digits):

Theorem A (the eight disguises). The eight corner twists $\theta \in \{0, \pi\}^3$ produce *identical* energy spectra — each is the untwisted problem seen through a relabeling of states. Practical payoff: anything requiring “all eight corners” needs *one* computer run.

Theorem B (average = mask). The sophisticated “character average” over the eight corners — the operator-level averaging done right — is *exactly* the parity version of the mask. Verified entry by entry: zero mismatches out of 8100. The fancy procedure and the simple deletion are the same object, and the deletion is eight times cheaper.

Theorem C (the general dictionary). Any twist whatsoever factorizes into (a relabeling) $\times$ (a genuine flux). This explains every entry of the old “twists keep half” table and closes the subject: twists and masks are one theory, and the mask is its strongest element.

One more exact selection rule fell out as a bonus: within the winding-free world, the uniform magnetization response at zero wavevector vanishes *identically* (measured: 5×10^{-30} , versus 0.333 for naive ED). Every bit of that signal in a naive simulation is artifact.

In plain words

Count, for each quantum process, how far it shifts the magnetization pattern. Wrap-around processes shift it by half a box; real processes shift it by zero. Delete the shifters. That single sentence is the method, it costs almost nothing, it provably deletes all of the artifact and none of the physics, and mathematics guarantees nothing sharper is possible.

7 A toolkit interlude: folding the universe into a small matrix

Everything so far treated the ice manifold perturbatively — valid when $|J_{\pm}|$ is small. The campaign’s second act pushes into *strong* coupling, and for that we need one more tool, the **Feshbach map** (also called a Schur complement). Despite the name it is one idea:

You care about a few states (the 90 ice states, call them P). The other 12 780 states (the spinon states, call them Q) matter only as temporary stopovers. Fold their effect into a corrected amplitude table for the 90, and solve the small problem.

The folded table is

$$F(z) = \underbrace{PHP}_{\text{direct}} + \underbrace{PHQ \frac{1}{z - QHQ} QHP}_{\text{out to } Q \text{ and back}}, \quad (6)$$

where z is the energy at which you ask the question. The exact energies of the full problem below the spinon continuum are precisely the self-consistent solutions of $E = \text{eig } F(E)$: the answer feeds back into the question. Nothing is approximated — F at the right z reproduces exact eigenvalues of the full H to machine precision.

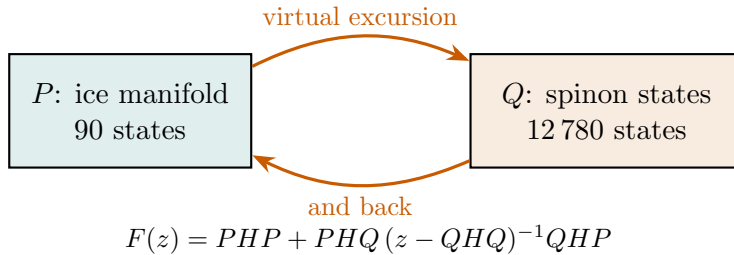


Figure 6: The Feshbach idea: excursions into the huge spinon space are summed exactly into a corrected 90×90 table. No frames, no overlaps, no approximation — and it exists at *any* coupling strength.

Two more characters:

Z , the ice weight (also: residue, purity). The fraction of a state that actually lives in P .

It is a property of each exact eigenstate $|n\rangle$ of the full H , not of the map:

$$Z_n = \langle n|P|n\rangle = \|P|n\rangle\|^2 = \sum_{c \in \text{two-in-two-out}} |\langle c|n\rangle|^2 \in [0, 1].$$

Be precise about what is being projected onto. The sum runs over the 90 *ice configurations* $|c\rangle$ — the spin patterns obeying the ice rule on every tetrahedron — not over Ising configurations in general: those form a complete basis, so projecting onto all of them is the identity and every Z_n would be 1. P is the rank-90 projector onto the span of that *subset*, $P = \sum_{c \in \text{two-in-two-out}} |c\rangle\langle c|$, diagonal in the Ising basis and purely combinatorial (Sec. 6.3); its complement Q is spanned by the other 12 780 configurations, each of which has at least one tetrahedron violating the rule, i.e. at least one spinon pair. So Z_n is the total probability that a measurement in the Ising basis returns an ice-obeying pattern. $Z_n = 1$ is a pure ice state, $Z_n = 0$ a state with no ice content at all, and $1 - Z_n$ is the *spinon dressing*: the weight the state has borrowed from Q by breaking ice rules virtually. One corollary worth flagging early: because the ice manifold is a set of configurations in the *local-z* frame, Z is defined relative to that frame — the same physical state has a different ice weight with respect to an x -basis ice manifold. Far from the Heisenberg point this is harmless (the z frame is the physical one); at it, it is the whole story (Sec. 14).

Two properties make Z more than a diagnostic. First, it is obtainable *without* eigenvectors: at a self-consistent root E_n of the branch $\lambda_n(z)$ of $F(z)$, $Z_n = [1 - \lambda'_n(E_n)]^{-1}$ — the flatter the folded map at a root, the purer the state (steep slope means the answer depended strongly on the asking-energy, i.e. on the excursions into Q). Second, it obeys an exact sum rule, $\sum_n Z_n = \text{Tr } P = 90$, the sum running over *all* 12 870 eigenstates: the ice manifold's 90 dimensions are shared out among the whole spectrum, and the shares always total 90. Sec. 9.5 turns that sum rule into a thermodynamic identity. At weak coupling $Z \approx 0.99$ (states are almost pure ice); by $J_{\pm} = -0.30$, $Z \approx 0.25$ — states are mostly spinon dressing. $1 - Z$ controls every error and every convergence rate in what follows.

The mask, again. The transport label is defined on ice states, so the mask applies to $F(z)$ verbatim: delete entries of F between different sectors. This “masked Feshbach map” is the campaign's precision instrument: *winding-free and valid at any coupling*.

Worked example you can check by hand

A three-level miniature contains the whole method. Take states $|1\rangle, |2\rangle$ (our “ice states”, in *different* transport sectors) and $|3\rangle$ (the “spinon”), with flip strength $t = 0.2$ into $|3\rangle$ and spinon cost $D = 1$:

$$H = \begin{pmatrix} 0 & 0 & t \\ 0 & 0 & t \\ t & t & D \end{pmatrix}.$$

Fold: excursions $1 \rightarrow 3 \rightarrow 2$ give $F(z) = \begin{pmatrix} f & f \\ f & f \end{pmatrix}$ with $f = -t^2/(D - z)$. The *off-diagonal* f connects different sectors: it is the analogue of a winding amplitude, generated purely by passing through the spinon. **Solve self-consistently:** $z = 2f(z)$ gives $z_0 = \frac{1}{2}(D - \sqrt{D^2 + 8t^2}) = -0.074456$, which is exactly the ground state of the full 3×3 matrix — fold-and-solve is exact. **Mask:** delete the off-diagonal; each sector alone solves $z = f(z)$, giving $z = \frac{1}{2}(1 - \sqrt{1.16}) = -0.038517$, twice (exactly degenerate) — the winding-free answer. Note it is roughly *half* the raw binding: on the real cluster, too, about half of the naive ground-state depth is artifact. **Counterterm** (Section 9): add $+0.037228$ to cancel the off-diagonal at z_0 ; the resulting honest Hamiltonian gives levels -0.039716 and -0.037228 — centred on the exact -0.038517 but split by 0.0025 . That split is the “residual z -dependence”: the counterterm cancels the artifact perfectly at one energy and imperfectly nearby. The purity here is $Z = 0.9352$; the split grows as $1 - Z$. Every phenomenon of the real campaign appears in this 3×3 example.

8 Strong coupling: the band dissolves, and the map survives

8.1 What breaks, and when

As $|J_{\pm}|$ grows, the neat separation “cheap ice band far below expensive spinons” degrades. On the 16-site cell the ice-derived band stops being cleanly separated from the two-spinon continuum at $J_{\pm} \approx -0.185$. Every earlier construction that relied on identifying “the 90 ice-like eigenstates” (so-called frames) fails there — and the campaign showed all of them fail *together*, because every one of their health diagnostics secretly measures the same quantity: the ice purity Z . There is no cleverer frame; the concept of “pick the ice-like eigenvectors” itself runs out.

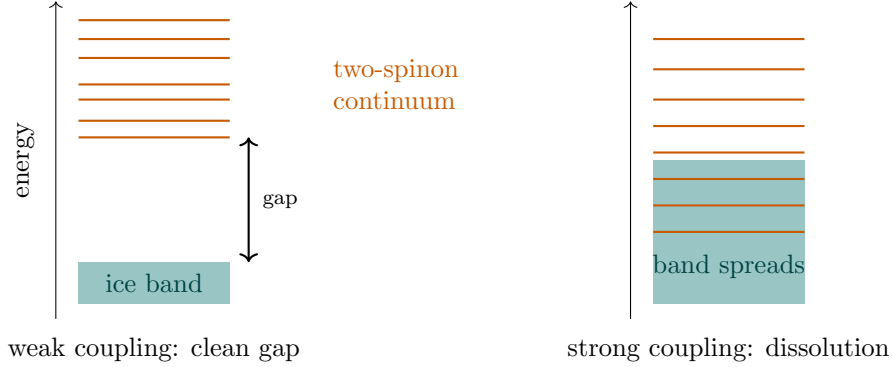


Figure 7: Weak coupling (left): the ice-derived band sits far below the spinon continuum. Strong coupling (right): the band widens and the continuum descends until they interpenetrate. Constructions that must *identify* band states die here; the Feshbach map, which never identifies anything, does not.

8.2 What survives

The masked Feshbach map does not care: it is built from the ice *configurations* (which exist at any coupling — they are just spin patterns), not from eigenvectors. Its self-consistent roots below the continuum edge remain exact, winding-free energies at every coupling tested. Highlights of the strong-coupling reconnaissance on the 16-site cell:

- **Completeness:** all 90 winding-free levels exist below the continuum up to $J_{\pm} = -0.22$; at -0.24 to -0.26 one goes missing; at -0.28 , thirteen (77 of 90 remain). The missing ones did not die — see next item.
- **The escaped levels are not lost.** They reappear *above* the continuum edge with exactly zero decay width, still tracked cleanly by the masked map. How that is possible — and when it stops being possible — deserves its own subsection (Sec. 8.3).
- **The flux law is restored.** The naive ground state gets the sign of the hexagon flux wrong at every coupling; the winding-free ground multiplet gives the correct π -flux sign throughout, fading smoothly as coupling grows ($\langle \text{hex} \rangle = -0.248$ at -0.02 down to -0.067 at -0.30).
- **Nothing dramatic happens to the raw ground state:** it evolves smoothly to $J_{\pm} = -1$ with no level crossing. The drama — band dissolution — is real physics of the *excitations*, and it is continuous, not a wall. (Its interpretation as a Higgs-type frontier of the emergent gauge theory is a separate, ongoing project.)

8.3 Life inside the continuum: tracking without repairing

What does it actually mean for a gauge-band level to “enter the spinon continuum”? The masked map answers this with unusual precision, because it is not one map but twenty-five: one independent block per transport sector, and *each block has its own continuum* — the spinon states that particular sector can reach. The global “continuum edge” quoted in dissolution plots is just the lowest of these.

So when a level of sector A rises above the *global* edge, the first question is: whose continuum is it swimming in? If the spinon states at that energy connect to *other* sectors, the level of sector A cannot decay into them — the transport label forbids it — and sector A ’s own map still has a perfectly ordinary, real, self-consistent root there. The level is a **bound state in the continuum** (BIC): embedded in the spectrum, yet with decay width exactly zero. This is not a hope but a measurement: the multiplets that escaped the band at strong coupling were recovered this way (for example a twelve-fold multiplet sitting $0.16 J_{zz}$ above the edge, width zero to machine precision). This is why the winding-free photon never decays in the regime we mapped, and why the “band merging” seen in naive ED there is frame artifact rather than a physical lifetime.

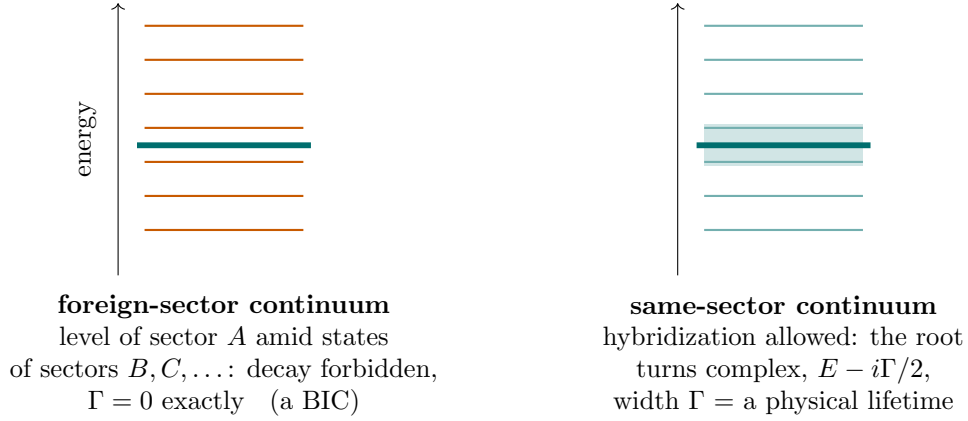


Figure 8: Two very different ways to sit inside a continuum. Left: the surrounding continuum belongs to other transport sectors (orange); the masked map’s per-sector root stays real and the state cannot decay — the mask keeps tracking it effortlessly. Right: the continuum in the state’s *own* sector has reached it; the real root disappears and the honest object is a complex “Gamow” root whose imaginary part is a genuine decay rate — physics to be reported, not an artifact to be repaired.

The mechanism has a genuine failure mode, and it is the physically interesting one. When the continuum *in the state’s own sector* reaches the level, nothing forbids hybridization: the real self-consistent root ceases to exist, and the correct object becomes a complex root $E - i\Gamma/2$ (a “Gamow” resonance). At that point the width Γ is not an error to be subtracted — it is a prediction: a finite photon lifetime, if the infinite system truly has one, must be allowed to appear exactly here. The mask has not lost track of the state; it has changed *what kind of object it reports*, from a level to a resonance.

Two honesty caveats keep this from being too tidy:

- On a finite cluster the “continuum” is itself a handful of discrete levels, so same-sector contact shows up as avoided crossings: the band state’s identity smears over several eigenstates, and tracking it becomes a matter of following *spectral weight*, not a single number.
- Whether a level is inside or outside the continuum depends on where the edge sits — and on a small cluster the edge is one of the *worst*-converged quantities there is. A verdict

of “escaped” can be an artifact of a misplaced edge. This is precisely the itch that the continuum-densification pillar of the next campaign (Sec. 13) scratches.

Lesson learned (the hard way)

The unconverged fixed point. Solving $E = \text{eig } F(E)$ by simple repetition (guess E , compute, feed back) converges at rate $1 - Z$ per round. At strong coupling $1 - Z \approx 0.5$, so three rounds leave a residual of about 10^{-2} — which was, for one embarrassing week, misread as a new physical effect (an “anchoring residue”, complete with a spurious extra peak in $C(T)$). A proper root-finder (a secant step) reaches 10^{-13} in 5–11 solves, and the “effect” evaporated: the ground multiplet is degenerate to 10^{-12} , the spurious peak gone. Moral: before interpreting a small number as physics, converge it.

Lesson learned (the hard way)

The physical third peak. After the spurious peak was removed, a *different* extra bump in $C(T)$ at $|J_{\pm}| \geq 0.22$ survived every convergence test — because it is real: a Schottky anomaly from a genuine gap inside the winding-free spectrum. At weak coupling it coincides with the ring bump and hides; at strong coupling they separate. One artifact peak and one physical peak had been conflated. Moral: not every surprise is a bug; the same rigor that kills artifacts must be allowed to certify physics. (The story goes deeper: “the third peak” turned out to name *three* distinct low-temperature bumps — one fake, two real. The full taxonomy, separated cleanly by the fold data, is Sec. 9.6.)

9 One honest Hamiltonian: the counterterm, its limits, the splice

9.1 Why there is nothing to cut: the ice block of H is empty

The masked map is exact but speaks only about the ice-derived band. For full thermodynamics — both peaks of $C(T)$, entropy, flux on the true dressed ground state — one wants a single honest Hamiltonian on the *whole* space. The natural first idea is pure surgery: take the full H , find the winding matrix elements between ice states, delete them, and leave everything else connected. No counterterm, no anchor — just a scalpel.

This fails for a startling reason: **there are no such matrix elements.** The ice block of the bare Hamiltonian, PHP , is *diagonal*. A single pair-flip acting on an ice state always breaks the ice rule on the tetrahedra it touches — it necessarily creates spinons and lands in Q . No ice state connects to any other ice state directly. Both the physical hexagon amplitude g_6 and the winding artifact g_4 are emergent: assembled from multi-step paths that leave the ice manifold, wander through spinon space, and return. They live inside the resolvent $(z - QHQ)^{-1}$, smeared across the geometry of Q — not in any entry of H a scalpel could reach. The 3×3 miniature of Sec. 7 shows it nakedly: the offending amplitude is $-t^2/(D - z)$, and no entry of the original matrix contains it; the matrix holds only t and D .

This is why deletion (the mask) operates on $F(z)$ — where the emergent amplitudes have *become* explicit matrix elements — and can never be performed on H itself. On H , the only possible surgery is **addition**: insert a new ice-block operator whose emergent effect cancels the winding paths. The counterterm is not a clever choice among alternatives; it is the only move the structure of H permits. And note that the “reconnection to the full Hamiltonian” comes free: the added operator touches only the 90×90 ice block, while PHQ , QHP , and QHQ remain byte-for-byte intact — spinons, dressing, and the upper band still flow through the untouched couplings.

9.2 The construction

The campaign’s construction, forced by the above: compute the masked-out (artifact) part of F at the self-consistent ground energy z_0 , and *subtract it from H* as a counterterm C supported purely inside the ice block:

$$H + C, \quad C = -(1 - \Delta) F(z_0) \quad (\Delta = \text{the mask}).$$

Because C lives inside the ice block, it does not disturb the spinon sector at all, and one ordinary diagonalization of $H + C$ delivers the entire 65 536-level spectrum, winding-free through third order in every loop class at once. This is the object behind the money plot of the campaign (the specific-heat scan of Section 11).

9.3 The obstruction theorem, and honesty about the residue

Can the counterterm be made *exactly* winding-free at all energies? **No — provably.** The transport label is only defined on ice states; a single spinon hop moves the polarization by a quarter of a box period, so no rule on the full space can conserve the label. Consequently $H + C$ carries an irreducible residual error that grows with the dressing: about 0.4% of the bandwidth at $J_{\pm} = -0.02$, 7% at -0.10 , 22% at -0.30 — the 3×3 miniature’s level split, writ large. The division of labor is therefore:

question	instrument	why
band energies, ring scale, flux	masked Feshbach roots	exact at any coupling
full $C(T)$, entropy, upper peak	$H + C$ spectrum	only full-space object
best of both	the splice	see below

The splice replaces the lowest 90 levels of the $H + C$ spectrum by the exact masked roots. Wherever the band is complete ($|J_{\pm}| \lesssim 0.22$ on 16 sites) the result is clean: exactly two peaks in $C(T)$, correct scales, no artifact.

9.4 What the counterterm cannot do

A natural hope: when band levels escape into the continuum (Sec. 8.3), perhaps $H + C$ “replaces” them too, carrying the correction into the dissolution regime. It does not — doubly, and provably.

First, C is **static and anchored**: it is built from F evaluated at one energy, the ground-state anchor z_0 . The states that run into the continuum are precisely the ones *farthest* from the anchor and the most spinon-dressed — so the cancellation is at its *worst* exactly for them. That is the $1 - Z$ residue of the previous subsection, seen from the other side.

Second, **the splice cannot rescue them**. The splice presumes a one-to-one assignment between “the lowest 90 levels of $H + C$ ” and “the 90 masked roots”. Once band levels and continuum levels interleave, that assignment simply does not exist: an escaped masked root sits *among* continuum eigenvalues of $H + C$, and there is no principled rule for which one it replaces. This — not numerical difficulty — is why the splice is certified clean only up to $|J_{\pm}| \approx 0.22$ on this cluster.

Nor can a smarter counterterm be built. Making C follow the escaped states would require either an energy-*dependent* $C(z)$ — but then $H + C$ is no longer a Hamiltonian and its thermodynamics is ill-defined — or a transport projection on the full space, which the obstruction theorem of the previous subsection forbids exactly (the spinon hop moves the label by a quarter period; no Hermitian operator conserves it). The wall is structural.

So in the dissolution regime the division of labor genuinely shifts: the per-sector masked map — with its BICs and Gamow widths — is the *only* exact instrument for band physics,

while $H + C$ degrades to a gross thermodynamic estimate whose growing residue must be quoted alongside it. Whether one can do better — dropping the counterterm altogether — is the subject of the next subsection.

9.5 A counterterm-free future: folding from both sides

The Feshbach fold of Sec. 7 has a mirror image. Just as $F(z)$ folds the spinon space into a corrected ice-block table, one can fold the ice space into a corrected *spinon*-block table:

$$G(z) = QHQ + QHP \frac{1}{z - PHP} PHQ.$$

Two facts make this mirror unusually pleasant. First, by Sec. 9.1 the block PHP is diagonal, so the fold-back factor $(z - PHP)^{-1}$ is trivial — $G(z)$ is just QHQ plus a cheap, low-rank correction. Second, and centrally: every eigenstate of H is seen by *both* maps, each in proportion to where the state lives. A level with ice weight Z_n — the quantity of Sec. 7, $Z_n = \langle n|P|n \rangle$, the fraction of that eigenstate sitting inside the ice manifold, with $1 - Z_n$ its spinon dressing — appears among the roots of F carrying weight Z_n , and among the roots of G carrying weight $1 - Z_n$ — and these weights are computable from the maps themselves (from the slope of F or G at the root). Summing both sides therefore counts every level *exactly once*:

$$\text{Tr } e^{-\beta H} = \sum_{\text{roots of } F} Z_n e^{-\beta E_n} + \sum_{\text{roots of } G} (1 - Z_n) e^{-\beta E_n}.$$

In plain words

Each energy level splits its vote between the two folded maps according to where the state actually lives: an almost-pure ice state votes mostly through the ice-side map, an almost-pure spinon state mostly through the spinon-side map, and the votes always add up to exactly one. No matching, no assignment, no double counting — the weights do the bookkeeping automatically, which is precisely what the splice’s one-to-one assignment could not do once band and continuum interleave.

The obvious winding-free version of this identity — keep the G -side sum *raw* (the continuum is physics, not artifact) and replace the F -side sum by the masked roots with their residues $\tilde{Z}_m$ — was built and measured. **It fails, instructively.** At $J_{\pm} = -0.05$, where every definition must agree, the naive weighted union put *zero* heat capacity at the ring peak ($C = 0.000$ at $0.84 g_6$, where splice and $H + C$ both say 0.157) and grew a spurious bump of height 0.65 at the *artifact* scale — taller than raw ED’s own artifact bump. The mechanism is worth engraving: the raw band states enter the union with small weight $1 - Z_n \approx 0.06$, but at their *raw absolute energies*, which sit a full artifact binding ($\sim g_4$) *below* the masked band — the depth the mask exists to remove. At $T \sim g_6 \ll g_4$ the Boltzmann factor $e^{+\beta \Delta}$ overwhelms any weight: the tiny remnant hijacks the entire low-temperature ensemble, and no weighting convention can save a scheme that leaves a lower-lying level in place.

Lesson learned (the hard way)

The hijacked ensemble. Weights cannot fix positions. In a thermal mixture, a state at lower absolute energy wins the $T \rightarrow 0$ limit no matter how small its weight — $e^{+\beta \Delta}$ beats any bookkeeping. A spectral-substitution scheme must therefore *remove* the levels it replaces, not merely down-weight them. (This is also the correct lens on the earlier “match-trace” trap: mixing spectra of two references makes their relative absolute placement load-bearing.)

The repair keeps both folds but changes what the mirror side supplies: **an identifier, not a weight**. Each raw eigenstate carries its exactly computable ice weight Z_n ; rank the raw levels by Z_n , remove the ninety highest — *wherever they sit*, interleaved with the continuum included, which is precisely the case the positional splice cannot handle — and insert the ninety masked roots at weight one:

$$\text{spectrum}_{\text{wf}} = \{ \text{raw levels} \} \setminus \{ 90 \text{ highest-} Z_n \} \cup \{ \text{masked roots} \}.$$

Counts are exact by construction; the masked roots are absolute (they share the spinon continuum’s energy axis, so no trace matching); BICs slot in wherever they lie. The entire scheme is raw ED + masked roots + bookkeeping — all cached campaign artifacts — and it scales to 32 sites, where the raw side goes stochastic and the band side uses the graded-truncation masked map.

Measured, on the 16-site cell across $J_{\pm} = -0.05$ to -0.40 :

- **Validation exact.** At -0.05 the fold’s ring peak lands at $0.84 g_6$, identical to splice and $H + C$ to the grid resolution (three-way spread $0.000 g_6$); the trace identity $\sum_n Z_n = 90$ holds to ten digits at every coupling.
- **The classification is measurable.** The ice weights are bimodal at weak coupling (band $0.89\text{--}0.97$ versus continuum ≤ 0.083 : an enormous Z -rank gap), and the gap closes by -0.20 — but at every computed coupling it closes *on an energy-degenerate multiplet* (cut energy split = 0 to machine precision), where the choice of which members to remove is thermodynamically invisible. The scheme stays well-defined deep into dissolution, and the Z -rank gap is the honest, continuous dial of how much longer that can last.
- **The ambiguity, finally measured.** At -0.30 the three definitions (counterterm, splice, fold) differ by 20–30% in $C(T)$ through the band region — the first quantitative measurement of the irreducible definitional ambiguity the obstruction theorem predicts. At -0.05 that spread is zero; it grows continuously with $1 - Z$.

Figure 9 condenses the whole scan into the peak positions. One reading caution that cost an afternoon: the campaign’s two-window peak extractor silently clips whichever maximum crosses its $T = 0.05$ split (strong-coupling raw “peaks” all read exactly 0.0497 in the archived records); peak positions must be re-extracted as true local maxima of the curves. The ruler, audited again.

One thing this scheme does *not* do is beat the obstruction theorem — nothing can. The theorem’s thermodynamic meaning is that “winding-free” is only exactly defined *on the band*; any extension to the full spectrum must decide which ninety full-space levels the band replaces, and that decision is a convention of order $1 - Z$. The counterterm, the splice, and the fold are therefore three *definitions* of the same non-unique reference. They provably agree at weak coupling (measured: exactly); their mutual spread at strong coupling is not a defect of any one of them but the honest, irreducible error bar on the concept of winding-free thermodynamics itself — and quoting that spread is the right practice. The fold simply has the best structural properties of the three: no anchor energy, no *positional* assignment (only a Z -ranking whose sharpness is itself measured), smooth behavior through dissolution, and a path to larger clusters.

For completeness, the genuinely different roads, with their price tags:

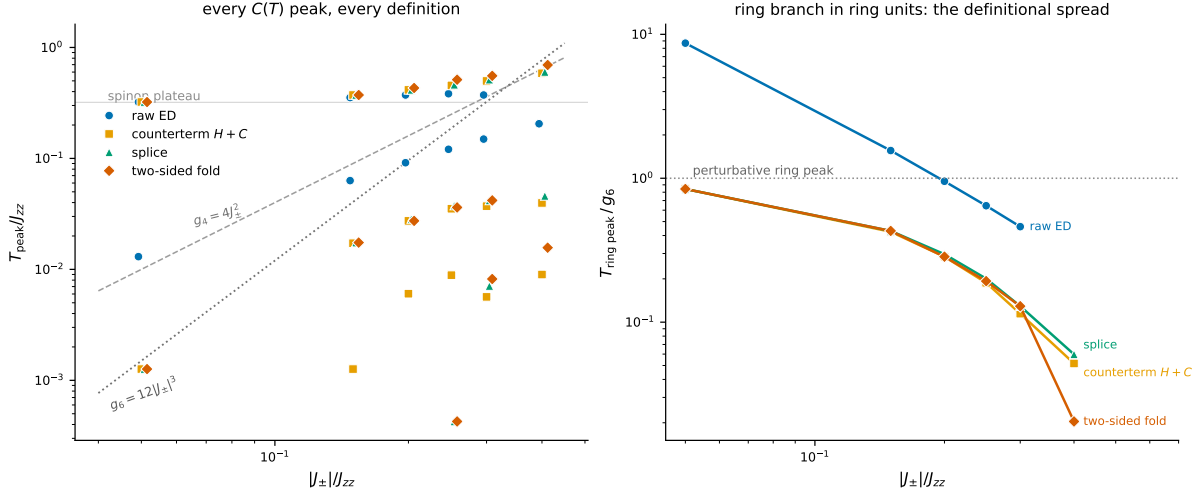


Figure 9: Every specific-heat maximum of every definition across the coupling scan (left) and the ring branch in units of g_6 (right). Three branches organize the left panel: the spinon anomaly on top, the ring branch below it, and a low-temperature third-peak branch that only the winding-free definitions possess (Sec. 9.6). On the ring branch, raw ED runs along the artifact scale g_4 and crosses $1 g_6$ near -0.22 purely by coincidence of scales; the three winding-free definitions collapse onto a single curve through -0.30 ($\sim 12\%$ spread in position) and fan out by a factor of three at -0.40 (fold $0.020 g_6$, counterterm 0.052 , splice 0.060) — the definitional ambiguity made visible. At -0.40 raw ED has no low-temperature peak at all: the artifact scale has grown into the spinon scale and swallowed every gauge-sector feature, while the winding-free definitions still resolve structure there.

route	what it buys	what it costs
counterterm $H + C$	complete spectrum from one solve	anchored at z_0 ; residue grows as $1 - Z$
splice	exact band + full continuum	needs the band below the continuum ($ J_{\pm} \lesssim 0.22$)
two-sided fold (Z -ranked)	no anchor, no positional assignment, BICs slot in, survives dissolution, scalable	$O(1 - Z)$ definitional ambiguity (irreducible for anyone); the naive <i>weighted</i> version fails outright (hijacked ensemble)
NLCE on open clusters	winding loops <i>never exist</i> — no torus, no artifact, by construction	converges only down to intermediate T ; the photon regime needs embedding orders far out of reach
girth-engineered cells	demotes the artifact itself to higher order in $J_{\pm}$	combinatorial search; the remedy shrinks the disease rather than curing it

The NLCE row deserves one sentence of emphasis: linked-cluster expansions sum *open* clusters, which have no glued edges and therefore no winding loops at all — the artifact this entire document fights simply never appears. That makes NLCE (a QED-backed tier of which already exists in the parent codebase) the one fully independent cross-check of winding-free thermodynamics in the temperature window where it converges; it cannot reach the photon scale, but agreement in the overlap window is evidence money cannot buy. Perturbative subtraction of the artifact from $\ln Z$ order by order is also possible but adds nothing: it works exactly where the counterterm already works, and fails where it fails.

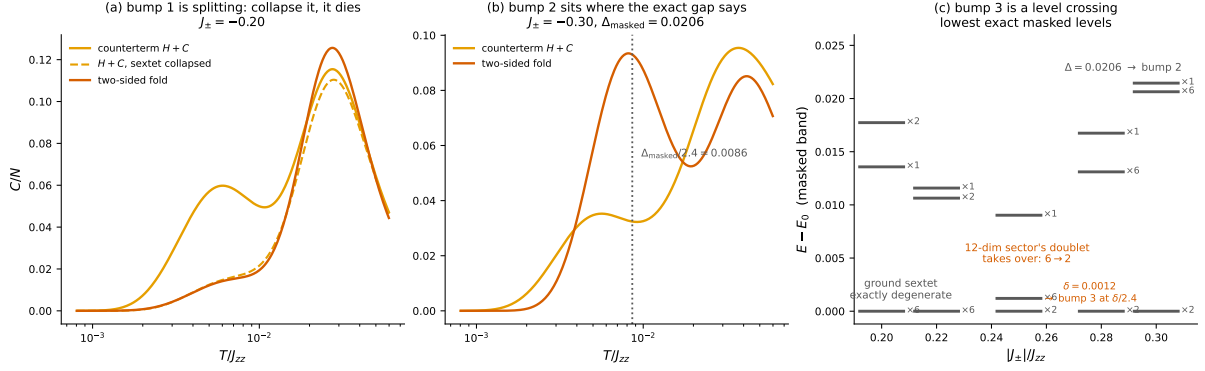


Figure 10: The third-peak taxonomy, demonstrated. (a) At $J_{\pm} = -0.20$ the counterterm curve has a low-temperature bump the fold lacks; collapsing the six lowest $H + C$ levels — the image of the exactly-degenerate masked ground sextet — to their mean (dashed) kills the bump and lands on the fold curve: bump 1 is pure multiplet splitting, the counterterm’s residue radiating. (b) At $J_{\pm} = -0.30$ the fold curve peaks at the two-level Schottky position $\Delta_{\text{masked}}/2.4 = 0.0086$ predicted by the exact masked internal gap $\Delta = 0.0206$: bump 2 is a certified gap speaking. (c) The lowest exact masked levels versus coupling: the ground sextet is degenerate to all digits through -0.22 ; at -0.25 the twelve-dimensional sector’s doublet dives below it ($6 \rightarrow 2$), leaving the splitting $\delta = 0.0012$ whose Schottky at $\delta/2.4 \approx 5 \times 10^{-4}$ is bump 3.

9.6 The third peak: three bumps, one name

Below the ring anomaly, $C(T)$ at strong coupling grows extra low-temperature structure — the “third peak” of Sec. 8’s lesson box and the lowest branch of Fig. 9. Asking *is it physical?* turned out to have three different answers, because “the third peak” named three different objects. Untangling them is a case study in the campaign’s central habit: the exact masked spectrum is the referee. A degeneracy it declares exact must not radiate a heat-capacity bump; a gap it certifies must. Figure 10 demonstrates all three verdicts directly from the computed spectra.

Bump 1: the counterterm’s splitting bump — artifact (Fig. 10a). In the exact masked map the sector ground states are *exactly* degenerate: six states, one per eight-dimensional transport sector, with spread 0.00000. The counterterm splits them — C cancels transport inside the ice block at one energy while PHQ keeps transporting — by a fraction of the bandwidth growing from 0.011 at -0.10 to 0.115 at -0.30 , and the split multiplet radiates a spurious Schottky bump. These are the counterterm-*only* peaks of Fig. 9 (at $T = 0.0013$ at -0.15 , 0.006 at -0.20), absent from splice and fold, whose lowest levels are the exact masked roots. The surgical proof: keeping only the five lowest $H + C$ levels reproduces the bump (0.00548 vs 0.00564 at -0.30); collapsing their splitting to zero removes it exactly. Verdict: **not physical** — it is the $1 - Z$ residue of Sec. 9.4 made visible at low temperature.

Bump 2: the internal-gap Schottky — physical (Fig. 10b). The exact winding-free spectrum has a genuine gap between the mask-protected ground multiplet and the seventh band level. Two independent constructions see the same number: at -0.22 the exact masked gap is 0.010634 and the counterterm’s is 0.010730. At weak coupling the resulting bump sits at $0.79 g_6$, indistinguishable from the ring anomaly at $0.84 g_6$ — it *hides inside the ring peak*; by -0.22 the two have separated by a factor of seven and it becomes a visible third maximum. At -0.30 the gap is 0.0206 and the fold and splice curves peak at $T \approx 0.008 \approx \Delta/2.4$, the textbook two-level Schottky position — and the fold’s

lowest levels there *are* the masked roots verbatim, so the feature is exact by construction.
Verdict: **physical**.

Bump 3: the multiplicity-change peak — physical, and a free diagnostic (Fig. 10c).

At -0.25 the ground multiplicity of the masked band drops from six to two: the twelve-dimensional sector takes over the band bottom, a level crossing *within* the winding-free ground manifold. The tiny exact splitting it leaves behind radiates the ultra-low maximum at $T = 4 \times 10^{-4}$ in the splice and fold curves at -0.25 (Fig. 9, lowest points).
Verdict: **physical** — and useful: a heat-capacity fingerprint of ground-sector reorganization, visible without ever inspecting an eigenvector.

Two honesty riders on the word “physical”. First, it means: a true feature of the exact winding-free reference *on this cluster*. Whether the 16-site band’s internal gap survives on FCC-32 — sectors forty times larger, flux unpinned — is exactly the kind of question the cluster, not the method, decides (Sec. 12). This rider has since been cashed for bump 3: the FCC-32 scan (Sec. 12.1) finds *no* multiplicity change at any coupling, so the crossing — and its little Schottky — are real physics of the 16-site torus and of nothing larger (Sec. 9.7 draws the moral). Second, at the couplings where the third peak stands separate, the definitional spread of Fig. 9 is already growing; the peak’s *existence* is definition-independent (its levels are sub-continuum masked roots, exact and shared by every construction that inherits them), but any statement about its precise *lineshape* carries the spread.

In plain words

Same temperature window, three causes. A counterterm error split an exact degeneracy — fake, and diagnosed because the exact masked spectrum says those states are degenerate to all digits. A real gap inside the clean band — real, certified by two independent constructions agreeing on the gap to three digits. And a change of which sector owns the ground state — real, and a bonus: the specific heat announces the level crossing on its own. The referee in every case is the same: exact degeneracies must stay silent, certified gaps must speak.

9.7 What the crossing means — and what it does not

Bump 3 records a change in the ground-state multiplicity, and a level crossing with a degeneracy change sounds like a phase transition. It is not one — and what it actually is deserves care, because it is arguably more interesting.

First, what it is not. It is *not* a ground-state level crossing of the actual Hamiltonian: the raw ground state of this cluster evolves smoothly and monotonically all the way to $J_{\pm} = -1$, with no crossing anywhere. The $6 \rightarrow 2$ crossing lives in the winding-free band, and it is a crossing between *different transport sectors* — the incoming doublet belongs to the twelve-dimensional polarization sector, displacing a sextet assembled from six eight-dimensional sectors. Transport sectors are not symmetry sectors of any local order parameter; they are the *topological* sectors of the emergent gauge field on the torus (polarization is electric flux threading the handles, Eq. 5). A change in which topological sector owns the ground state is a reordering of flux-sector energetics: no local observable jumps, nothing orders, no symmetry breaks. On a single finite cluster, that is spectroscopy, not a transition.

Here is why it is interesting anyway. In a gauge theory, the energy splittings between flux sectors are precisely the quantity whose *system-size scaling* diagnoses the phase (the ’t Hooft criterion — and the reason Sec. 13 insists the masked-out sectors be recycled, not discarded): in the photon phase the splittings must vanish as the torus grows; in a Higgs or confined regime

they saturate. The crossing is a single data point of exactly the right diagnostic, waiting for a size axis.

And it sits at a suspicious address. The crossing lands at $J_{\pm} = -0.25$ — the coupling where one line of theory (the self-consistent exclusion-boson treatment) predicts the π -flux spin-ice phase terminates, $J_{\pm}/J_{zz} = -1/4$; a competing estimate puts the terminus near -0.5 . It is also where the masked band’s completeness begins to degrade. So a tempting reading exists: *the 6→2 crossing is the finite-size shadow of the QSI terminus* — the gauge theory’s sector energetics inverting as the Higgs frontier approaches. Three caveats before believing it: cubic-16’s ground-state flux is pinned by pure counting, so its sector energetics are partly arithmetic; the splitting is tiny ($0.0012 J_{zz}$) and must sharpen, not vanish, with system size to mean anything; and the definitional spread is already growing there (though the crossing itself is between sub-continuum masked roots — exact and shared by every construction).

Is it still a spin liquid beyond? What theory requires: the deconfined phase is a genuine phase — the photon either exists or it does not — so a true $T = 0$ transition out of it must occur *somewhere* in the thermodynamic limit. Beyond it, though, no further sharp structure is required: the spinons carry fundamental gauge charge, so the Higgs and confined regimes are smoothly connected to each other (Fradkin–Shenker), consistent with the continuous dissolution we measure. What this cluster sees: every gauge signature fades smoothly — the flux expectation from -0.25 at weak coupling to -0.067 at -0.30 and -0.009 at -0.35 , the ring scale renormalized to $\sim 0.1 g_6$, the purity to $Z \approx 0.25$ — with no condensation, no symmetry breaking, no raw-level crossing. That is exactly what a true transition *rounded by a tiny torus* looks like, and also exactly what a crossover looks like; sixteen sites cannot tell them apart. Best current statement: the terminus lies somewhere in $[-0.25, -0.5]$, and this cluster cannot adjudicate.

Two measurements would decide it. (i) Does the crossing reproduce on FCC-32 — flux unpinned, sectors up to dimension 396 — and does its coupling converge with cluster size and truncation depth? A physical terminus shadow drifts systematically; an artifact wanders or vanishes. (ii) The flux-sector splitting scaling itself, across sizes and girths.

Measurement (i) has now been made (Sec. 12.1), and the answer is: it vanishes. On FCC-32 the band bottom belongs to the unique largest (zero-polarization) sector at *every* coupling from -0.05 to -0.32 , with a twelve-fold degenerate multiplet of 140-dimensional sectors above it and a gap that only grows — and deepening the truncation from $L = 1$ to $L = 2$ *entrenches* that ordering (the gap at -0.22 triples), the opposite of what a hidden crossing would do. This is exactly the ordering the photon phase predicts: flux costs energy, so the zero-flux sector sits lowest. The correct reading of cubic-16’s crossing is therefore deflationary: its pathological sector structure (flux pinned by counting, sectors of dimension at most twelve) put the *wrong* sectors on the bottom at weak coupling, and the celebrated 6→2 crossing at -0.25 was the small cluster relaxing into the correct ordering — a cluster artifact, not a terminus shadow. The coincidence with the $-1/4$ prediction was a coincidence. Bump 3 of the taxonomy is correspondingly a cubic-16-only feature, exactly the outcome its cluster rider anticipated.

What survives, sharpened: the terminus question itself now rests entirely on measurement (ii) — the sector splittings, measured on both clusters and growing smoothly with coupling, are the raw material of the ’t Hooft scaling analysis, and the recycled-sector program of Sec. 13 remains the road to it. And there is a methodological moral worth stating plainly: the machinery in this document not only found a sharp feature, it then *killed its own most exciting interpretation* with a controlled measurement on a better cluster. That is what the validation-first culture of this campaign is for.

Lesson learned (the hard way)

Repairs that are not repairs. Two tempting improvements were tested and made things *worse*. (i) Forcing the upper $H + C$ band to inherit the exact degeneracy pattern of the masked map (“sector symmetrisation”) increased the error everywhere — the pattern is exact but the eigenvectors are not, so the symmetrisation imposes structure the true operator does not have. (ii) Hand-built counterterms from a catalogue of loop shapes missed whole classes of winding processes and shifted the *physical* part of the map by up to 42%. Moral: a correction must be derived from the exact object (F itself), not from a plausible model of it.

Lesson learned (the hard way)

Estimator traps. Three measurement bugs each briefly masqueraded as physics. (i) A comparison routine silently *shifted* one spectrum to match the mean of another — invisible in level differences, corrupting every absolute-energy comparison. (ii) Classifying $C(T)$ peaks by Boltzmann weights mislabels them; the correct weight is each level’s share of the heat-capacity *variance*. (iii) The entropy sum rule $\int C/T dT = \ln 2$ converges like $1/T_{\max}^2$: truncating at $T_{\max} = 60$ leaves a deficit that looks exactly like a lost energy level and does not improve with grid refinement. ($T_{\max} = 600$ resolves it.) Moral: before blaming the physics or the method, audit the ruler.

10 Turning the second dial: what $J_{\pm\pm}$ actually does

Everything so far lived on the XXZ line, where the only transverse coupling is $J_{\pm}$. There the two anomalies of $C(T)$ sit absurdly far apart: at $J_{\pm} = -0.125$ the winding-free gauge peak is at $0.0121 J_{zz}$ and the spinon peak at $0.274 J_{zz}$, a ratio of 0.044 — a factor of twenty-three. The real materials refuse to behave: $\text{Ce}_2\text{Hf}_2\text{O}_7$ shows two heat-capacity anomalies at 65 and 24 mK, a ratio near 0.37. A factor of twenty-three against a factor of under three is not a discrepancy to be split; it is a different picture of the world.

The obvious suspect is the coupling we had switched off. Real cerium pyrochlores are XYZ, and in the $J_{\pm}$ language that means a second transverse term, the *pair flip* $J_{\pm\pm}$, which flips two spins the *same* way instead of exchanging them. It is not small in the fits: for the published $\text{Ce}_2\text{Hf}_2\text{O}_7$ parameters $J_{\pm\pm}$ is comparable to $J_{\pm}$ itself. So we turned the dial — an 8×7 grid, 56 cells, each one a complete exact band — and asked whether it closes the gap.

10.1 What a pair flip costs

Start with what the move does to the ice rules, because that sets everything else.

Worked example you can check by hand

Take a tetrahedron obeying two-in-two-out, so its charge is $Q_t = 0$, and recall the energy is $\frac{1}{2} \sum_t Q_t^2$ in units of J_{zz} .

An *exchange move* ($J_{\pm}$) flips one up and one down spin on a bond. The tetrahedron containing both keeps two ups: $Q_t = 0$, no cost. The two outer tetrahedra — one for each flipped site — gain and lose one respectively: $Q = \pm 1$, costing $\frac{1}{2}$ each. **Total: one unit.** That is a spinon–antispinon pair, the familiar excursion.

A *pair flip* ($J_{\pm\pm}$) flips two spins that were both pointing the same way. Now the shared tetrahedron goes from two ups to four: $Q_t = +2$, costing $\frac{1}{2}(2)^2 = 2$. The two outer

tetrahedra each gain one: $\frac{1}{2}$ each. **Total: three units.**

So a pair flip is three times as expensive as an exchange, and it is *qualitatively* different: it manufactures a **double monopole**, a tetrahedron carrying twice the elementary gauge charge, flanked by two ordinary spinons. It also fails to conserve the total S^z , which is why the calculations in this section need the full 65 536-state space rather than the 12 870-state sector the XXZ line lives in.

One thing does *not* change. The emptiness theorem of Sec. 9.1 survives verbatim: a pair flip acting on an ice state always breaks the ice rule, so *PHP* is still exactly diagonal — we checked it numerically, and it is zero to the last bit. $J_{\pm\pm}$ buys no new direct ice-to-ice matrix element. Whatever it does to the gauge physics, it does through the resolvent, like everything else.

10.2 Three measured effects, and a squeeze

With the complete band in hand at every cell, three things happen as $J_{\pm\pm}$ is turned up at fixed $J_{\pm}$. The numbers below are at $J_{\pm} = -0.125$, the coupling closest to the $\text{Ce}_2\text{Hf}_2\text{O}_7$ fits, as $J_{\pm\pm}$ runs from 0 to 0.30.

1. **The band sinks.** Its bottom drops from $-0.437 J_{zz}$ to $-0.822 J_{zz}$. This is ordinary variational bookkeeping: expensive as they are, virtual pair excursions are extra ways for the ice states to lower their energy, and more channels always means a deeper ground state.
2. **The band widens** — from $0.075 J_{zz}$ to $0.184 J_{zz}$, a factor of 2.5. The gauge peak tracks the bandwidth, so it rises: $0.0121 \rightarrow 0.0185 J_{zz}$, up 53%. The photon gets *faster*.
3. **The spinon scale softens** — the upper peak falls $0.274 \rightarrow 0.168 J_{zz}$, down 39%. Pair flips help the spinon continuum more than they help the band, because a region already full of monopoles has far more ways to use them, so the continuum descends faster than the band does and the gap between them closes from above.

The two peaks therefore approach each other *from both sides at once*: the gauge anomaly climbing, the spinon anomaly sinking. The ratio duly improves, $0.044 \rightarrow 0.110$.

In plain words

Think of the two peaks as two shelves, one very low and one high, and you are trying to bring them together. Turning up $J_{\pm\pm}$ raises the low shelf and lowers the high one simultaneously — exactly the move you wanted. It works. It just does not work anywhere near hard enough: the shelves close from a factor of twenty-three to a factor of nine, and the experiment wants a factor of three.

10.3 Why it stops: the dial destroys what it moves

The squeeze does not gently saturate — it terminates, and for a reason that is inevitable in hindsight.

The mechanism by which $J_{\pm\pm}$ closes the gap *is* the admixture of non-ice configurations into the band. The ice weight Z (Sec. 7: the fraction of a level living inside the ice manifold) measures exactly that admixture, and it decays as the dial turns: at $J_{\pm} = -0.125$ the median Z slips from 0.723 to 0.640; at $J_{\pm} = -0.20$, from 0.497 to 0.257. Push further and there is

nothing left to speak of. At $(J_{\pm}, J_{\pm\pm}) = (-0.05, 0.30)$ only 25 of the 90 levels survive as band states at all and the largest ice weight anywhere is 0.001 — the band has dissolved completely.

That corner is worth naming precisely, because it is the one place in the plane where dissolution is real rather than instrumental. It is not set by $|J_{\pm}|$, the dial that governs dissolution on the XXZ line. It is set by the *ratio*: the band dies when $J_{\pm\pm}/|J_{\pm}| \gtrsim 5$. At $J_{\pm} = -0.05$ that threshold is crossed by $J_{\pm\pm} = 0.25$ (marginal, 86/90) and comprehensively by 0.30; at $J_{\pm} = -0.20$, where the same $J_{\pm\pm} = 0.30$ is only 1.5 times the exchange, the band is intact at 90/90. When the pair term dominates the exchange term outright, the ice manifold stops being the right low-energy description — and no amount of careful bookkeeping about winding loops can rescue a band that is no longer there.

So the two effects are the same effect wearing different hats. You close the gap by dressing the band with monopoles; dress it enough and you have destroyed the band. The best ratio in the whole plane, 0.1216 at $(-0.16, 0.30)$, sits exactly where you would expect from that trade-off: at intermediate exchange, with the pair term pushed as far as the band will tolerate and not one step further.

10.4 The verdict, and one honest complication

The ceiling over all 56 cells is **0.1216**, against a measured 0.37. $J_{\pm\pm}$ moves the ratio by a factor of about 2.5 and needs a factor of eight. **On the 16-site cell, the second dial does not rescue the peak separation.** Combined with the cluster comparison of Sec. 12 — where FCC-32 pushes the ring peak *further* down, to $0.29 g_6$ against cubic-16's 0.84 — the honest reading is that the 24 mK anomaly in $\text{Ce}_2\text{Hf}_2\text{O}_7$ is probably not the photon peak at all.

The complication is a second kind of merging, and it is real. At strong exchange and large pair coupling — nine cells with $|J_{\pm}| \geq 0.20$ — the gauge peak and the sub-gauge peak of Sec. 9.6 run together into a single maximum. At $(-0.27, 0.20)$ there is one sub-spinon maximum at $0.0121 J_{zz}$ where both neighbours in $J_{\pm\pm}$ carry two, near 0.005 and 0.039. This is a genuine merge of two low-temperature features; it is *not* the gauge peak meeting the spinon peak, and it moves the ratio the wrong way, down to 0.028–0.045. Any convention that reports “the highest maximum below the spinon split” will report the merged position and produce a branch that appears to jump. It jumps because the peaks merged, not because the tracking failed.

Lesson learned (the hard way)

An earlier pass over this plane reported a ratio of 0.332 at $(-0.125, 0.30)$ — tantalisingly close to experiment, and wrong. The anchor solve had found only 27 of the 90 band levels there, and the two-sided fold removes exactly as many raw levels as it has replacements for. So 63 ice-descended raw levels stayed in the ensemble *beside* their own replacements, and the low-temperature peak was radiating from the double-counted remainder. With the complete band the same cell reads 0.110.

The diagnosis is worth keeping, because the failure was silent: every individual number was computed correctly. Two habits catch it. First, a fold that replaces a band must replace *all* of it or refuse to run — a partial replacement is not a weaker result, it is a different and meaningless one. Second, when an instrument is upgraded, re-run the cells where the old one *succeeded*: the bordered solver reproduces the old ratios to the digit wherever the old band was complete $((-0.20, 0.15): 0.0893 \rightarrow 0.0893)$, which is what makes the changes elsewhere trustworthy rather than merely different.

11 What the campaign established

A compact scoreboard of the physics results, all on the 16-site cell unless stated:

- **Naive ED never measures the photon.** Its low- T peak tracks the artifact scale g_4 (exponent $1.99 \approx 2$); against the photon scale it drifts by a factor 40. The winding-free peak tracks g_6 (exponent $3.10 \approx 3$) and approaches $1.02 g_6$ as $J_{\pm} \rightarrow 0$ — the perturbative prediction, recovered exactly where it must hold.
- **Flux-sign physics is real and restored:** π -flux suppression versus zero-flux enhancement of the ring scale (e.g. $0.73 g_6$ vs $1.13 g_6$ at $|J_{\pm}| = 0.046$), with the correct sign of the hexagon expectation at every coupling — naive ED gets the sign wrong everywhere.
- **The spinon peak is robust:** identical (4 digits) between naive and winding-free treatments up to $|J_{\pm}| \approx 0.07$ — as it must be, since spinons live outside the ice manifold. A method that “cleans” the spinon peak would be destroying physics; this one provably does not touch it.
- **The photon does not decay** in the winding-free world: all masked levels have zero width at every coupling tested; apparent band merging in naive ED is frame artifact. Escaped levels at $|J_{\pm}| \geq 0.24$ are bound states in the continuum, protected by transport symmetry.
- **A third, physical Schottky peak** exists at $|J_{\pm}| \geq 0.22$ from a genuine internal gap of the winding-free spectrum.
- **Application to $\text{Ce}_2\text{Hf}_2\text{O}_7$:** at literature couplings the naive ED peak lands deceptively close to the measured 24 mK anomaly — but the winding-free calculation shows that agreement is a coincidence between the *artifact* scale and the data; the true ring peak on this cell is at 6 mK. A cautionary tale for fitting experiments with naive small-cluster ED.
- **Dissolution is continuous:** band isolation is lost at $J_{\pm} \approx -0.185$, completeness degrades from -0.24 , and by -0.30 the band is as wide as the spinon scale. No sharp wall, no phase transition of the ground state — and the strong-coupling interpretation (Higgs frontier of the emergent gauge theory) is the subject of the follow-up project.

12 The cluster is the error bar

With the artifact provably removed, what limits accuracy? The box itself — but now in the ordinary, honest way: through what it can and cannot represent. The comparison between the two available cells is stark:

	cubic 16-site	FCC 32-site
ice states	90	2970
transport sectors	25	85
largest sector	12	396
flippable hexagons per state	0 or 4 only	0, 4, 6, 7, 8
ground-state flux	<i>pinned by counting</i>	free
winding-free ring peak at $J_{\pm} = -0.05$	$0.84 g_6$	$0.29 g_6$

A factor of *three* in the headline number, between two clusters with the identical, exact protocol. The 16-site cell is simply a poor host: its ice manifold is so small that the flux is fixed by arithmetic before any dynamics happens. This is the single most important systematic in the whole program, and it motivates everything in the final section.

12.1 First steps onto FCC-32: calibrate, then scale

The crossing question of Sec. 9.7 forces the move to FCC-32, and the move immediately raises a practical wall: the $S^z = 0$ block there has 601 million states. The Feshbach fold’s escape hatch is **graded truncation**: keep only the spinon states reachable from the ice manifold in at most L pair-flip moves. Physically, L is *how deep a virtual spinon excursion the calculation can follow before being forced home* — one freshly created adjacent pair at $L = 1$; the pair separated, or a second pair, at $L = 2$; and so on. Since strong-coupling dressing *is* spinons wandering far and often, L is the convergence dial of everything on the large cluster, and its ladder is the error bar.

Lesson learned (the hard way)

The instrument that failed its physical. A truncated FCC-32 band pipeline already existed (one-pair truncation, map evaluated at a single anchor energy), and the obvious plan was to simply run it near the crossing. Calibrating it first on cubic-16 — where the exact answer is known — stopped that: it shows *no crossing at all*, even at -0.30 , with multiplet gaps wrong by a factor of four. The fixed anchor misses the self-consistent $\sim Z$ renormalization of level spacings. Run uncalibrated, it would have returned a confident, wrong verdict on precisely the question at stake. Rule: an instrument earns the big cluster only by first reproducing known truth on the small one.

The instrument that passes: each sector’s masked spectrum is exactly the spectrum of the Hermitian *bordered matrix* $M_s = \begin{pmatrix} (PHP)_{ss} & (PHQ)_s \\ (QHP)_s & QHQ \end{pmatrix}$ (the same object that explained the BICs in Sec. 8.3), so sparse Lanczos on M_s delivers each sector’s ground energy with *no anchor and no self-consistency loop* — exact within the truncation. Its cubic-16 ladder, against the known crossing:

L	truncated space	cost	crossing verdict
1	1 242	instant	absent, even at -0.30 — fails
2	5 754	instant	present, displaced into $(-0.25, -0.30)$
3	11 130	seconds	present, within ~ 0.02 of the true coupling
4	12 870 (= full)	seconds	exact to all digits

Monotone convergence from below, with the crossing coupling drifting systematically toward truth — so on FCC-32 the drift between L levels is a measured error bar, not a hope. The FCC-32 costs, measured on the campaign machine: $L = 1$ has 140 442 states (seconds per sector); $L = 2$ has 2 511 450 states, a 50-million-element sparse Hamiltonian built in 12 seconds, and one sector ground in 154 seconds — about 3.6 hours per coupling for all 85 sectors; $L = 3$ fits in memory but costs roughly a day per coupling, reserved for single-point convergence checks.

The scans are done, and the result is unambiguous. FCC-32’s 2 970 ice states split into 85 sectors with dimensions 1×396 , 12×140 , 6×72 , 24×12 , 12×6 , 24×4 , 6×1 . At *every* coupling scanned ($L = 1$: thirteen points, -0.05 to -0.32 ; $L = 2$: four points), the band bottom belongs to the unique 396-dimensional zero-polarization sector, with the twelve 140-dimensional sectors forming an exactly degenerate runner-up multiplet:

gap to runner-up multiplet	-0.05	-0.22	-0.26	-0.30
$L = 1$	0.0014	0.0137	0.0164	0.0191
$L = 2$	0.0018	0.0374	0.0473	0.0573

No crossing, anywhere — and the truncation trend runs *opposite* to a hidden one: going deeper ($L = 1 \rightarrow 2$) makes the zero-polarization sector’s dominance stronger, where on cubic-16 going deeper is what *revealed* the crossing. The verdict this feeds is drawn in Sec. 9.7. (Measured runtimes for the record: 1.6–2.9 hours per $L = 2$ coupling — the weak-coupling points converge fastest.)

Honesty about what FCC-32 will and will not deliver. The sector-ground scan answers the crossing question. The full *peak* program needs the entire 2970-level winding-free band, and there the estimator ladder currently reads: full self-consistency — validated excellent on cubic-16 (ring peak to 0.3% at $L = 3$) — is unaffordable on FCC-32; the cheap single-anchor slope correction lands 7–14% high on the ring peak; and a naive multi-anchor version is *unstable* at strong coupling (anchors placed near the descended continuum corrupt the branch assignment — measured, not conjectured). A branch-tracked multi-anchor root finder is the one named engineering gap between here and the full peak scan on FCC-32. And two things never come from FCC-32 at all: the raw-ED comparison curve and the spinon anomaly, both of which need the 601-million-level spectrum — those remain cubic-16 deliverables, which is an acceptable division of labor since every peak that matters at strong coupling (ring, third, crossing) is band-derived.

13 The road ahead: beating finite size at strong coupling

The next campaign asks: *how do we maximally negate the remaining finite-size effects, in the strong-coupling regime, without introducing new physics or killing real physics?* The answer designed above rests on a diagnosis: at strong coupling the dominant box effect is no longer one thing but three, and each has exactly one clean remedy.

13.1 Residual 1: topology — already solved

The winding artifact itself is handled by the mask, which is combinatorial and coupling-independent: it works identically at $J_{\pm} = -0.30$ and -0.03 . Nothing new needed — but everything below must be built *on top of* the masked map, never instead of it.

13.2 Residual 2: matched winding pairs — solved by geometry

Processes that wind $+1$ then -1 (net zero) are topologically trivial: no label can see them, and the mask correctly keeps them. At weak coupling they are negligible (they cost two powers of the already-small winding amplitude); at strong coupling they are not. The *only* knob is geometry: their size is set by the length of the shortest winding loop the cell permits. Hence **girth engineering**: among all simulation cells of a given size (including skewed, non-rectangular ones), choose the cell that maximizes the shortest wrap-around ice loop.

Because the loop census is a counting exercise, thousands of candidate cell shapes can be screened without a single diagonalization, scoring each by: shortest winding loop (maximize), number of contractible hexagons (keep), flippability variety (keep), flux unpinned (require), largest transport sector (keep tractable).

13.3 Residual 3: the coarse spinon continuum — solved by twists, reborn

At strong coupling every low-energy state is heavily dressed by spinon excursions ($1 - Z$ up to 0.75), but the small box offers only a handful of discrete spinon levels where the infinite crystal has a dense continuum. This misplaces the continuum edge, exaggerates the sharpness of the dissolution, and expels the bound-in-continuum levels.

Here twist averaging returns — in the *opposite* role from Section 5. Threading a flux shifts the spinon levels; averaging over many twist values sweeps them across the true continuum,

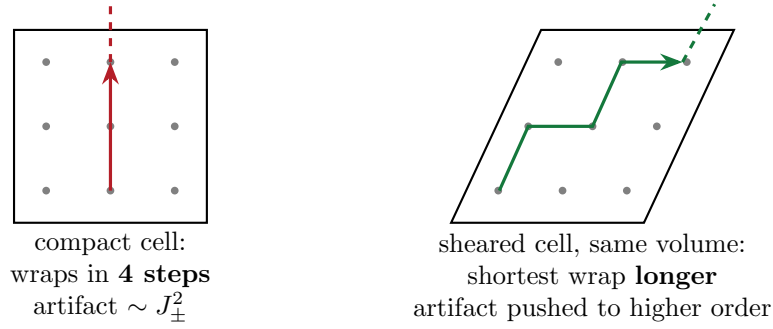


Figure 11: Girth engineering. The irreducible (net-zero-winding) artifact is controlled by the length of the shortest wrap-around loop. Skewing the simulation cell at fixed size can lengthen that loop, demoting the residual artifact by powers of $J_{\pm}$ — at zero cost in Hilbert-space size. The search over cell shapes is purely combinatorial (no diagonalization needed).

exactly the classic “boundary-condition integration” of electronic structure theory. Crucially, the mask is applied *at each twist* (the transport label does not depend on the twist), so topology removal and continuum densification never compete:

mask handles the loops; twists handle the levels.

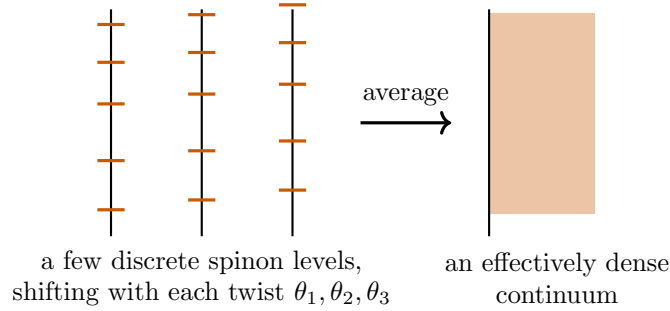


Figure 12: The twist reborn. Each twist value slides the discrete spinon levels; averaging the *masked* results over twists reconstructs a dense continuum for the virtual excursions — correct dressing, smooth continuum edge — without adding any new operator to the Hamiltonian. Each twisted problem is an exact microscopic Hamiltonian, so no physics is invented.

A free diagnostic comes along: genuine contractible physics is provably twist-independent, so the *spread* of the masked results across twists is a direct, honest error bar on whatever finite-size effect remains. And a second dividend: a trustworthy continuum edge is exactly what the bound-state-versus-resonance verdicts of Sec. 8.3 hinge on — on a coarse, undensified continuum, the question “did this level really escape?” is not even well-posed.

13.4 Honesty items

Three more commitments, so the method neither invents nor kills physics at strong coupling:

- **Graded truncation.** On the 32-site cell the spinon space must be truncated (one pair, two pairs of spinons). At strong coupling the one-pair truncation is not obviously sufficient; the protocol requires the one-pair/two-pair difference to be measured and quoted at every coupling.

- **Recover, don't repair.** Where winding-free levels escape into the continuum, they are recovered as symmetry-protected bound states (per transport sector), not patched with counterterms. Where the continuum in the *same* sector genuinely touches a level, its decay width is computed and reported as physics — at strong coupling a finite photon lifetime, if real, must be allowed to appear.
- **Recycle the removed sectors.** The masked-out winding sectors are not garbage: their energy splittings versus twist and cell size are precisely the flux-sector free energies whose scaling distinguishes the photon phase from a Higgs phase. The artifact we negate in the physical sector becomes the measurement in the diagnostic sector — one production run serves both this program and the strong-coupling phase-diagram project.

13.5 The validation ladder

Every stage of the new protocol must pass, in order:

1. **Weak-coupling anchor:** ring peak $\rightarrow g_6$ with exponent 3 as $J_{\pm} \rightarrow 0$ on the new cell (the exact perturbative limit).
2. **Spinon neutrality:** the spinon peak agrees with naive ED at weak coupling to 4 digits (the mask must not touch it).
3. **Flux law:** correct π -flux sign, and the known flux-asymmetry between $\pm J_{\pm}$ at matched magnitude.
4. **Physical features survive:** the third Schottky peak present; no feature removed that the exact masked reference possesses.
5. **Sum rule:** entropy integral returns $\ln 2$ (with the $T_{\max}$ lesson applied).
6. **Cross-size consistency:** dimensionless ratios (peak/ g_6 , gaps/ g_6) drift monotonically and modestly from $16 \rightarrow 32 \rightarrow$ girth-optimized cells, with twist-spread error bars quoted.

14 Where the method ends: the Heisenberg point, and beyond

Every method should be able to say where its own writ runs out. For the mask the answer is unusually crisp — and it is *not* at the band dissolution. Dissolution only changes which instrument carries the load (Secs. 8.3 and 9.4). The genuine boundary lies deeper in coupling, at a point of enhanced symmetry.

14.1 The Heisenberg point

Our transverse term is $-J_{\pm}(S^+S^- + S^-S^+) = -2J_{\pm}(S^xS^x + S^yS^y)$, so at

$$J_{\pm} = -\frac{1}{2} J_{zz}$$

the Hamiltonian becomes $J_{zz} \mathbf{S}_i \cdot \mathbf{S}_j$: the antiferromagnetic **Heisenberg** model, with full rotational (SU(2)) symmetry — no direction in spin space is special anymore. The campaign's reconnaissance stopped at $J_{\pm} = -0.30$, well short of this point, and the trend at the frontier was already steep: purity $Z \approx 0.25$ and falling, thirteen of ninety band levels escaped.

Two different things fail as this point is approached, and separating them is the whole insight:

1. **Quantitatively, nothing crashes.** $F(z)$ is linear algebra on spin patterns; it exists at any coupling, and every self-consistent root of the *unmasked* map is an exact eigenvalue of H no matter how small Z becomes. The machinery runs at -0.5 as happily as at -0.05 .
2. **Structurally, the mask breaks the symmetry.** The transport label is built on the ice manifold *in the S^z basis*. At the Heisenberg point the choice of z -axis is pure convention — so “the winding-free model” built on the z basis and the one built on the x basis are *different operators with different spectra*, disagreeing about which processes are artifacts. A cleaning procedure whose output depends on a convention is no longer describing the system.

In plain words

The mask does not remove “the” finite-size artifact; it removes the finite-size artifact *of a particular description* — the ice-gauge description in the z basis. Approaching the Heisenberg point, that description carries less and less of the low-energy weight, until at the point itself it becomes one of three equivalent conventions. The premise dissolves before the method does.

This failure is measurable, which turns it from a worry into a diagnostic: build the label on the z , x , and y axes in turn and record the **axis spread** of the masked observables. At weak coupling the anisotropy picks the axis physically and the spread is negligible; toward -0.5 it must grow, and its size *is* the meaninglessness of the mask, quantified. At the point itself the honest labels are total-spin quantum numbers (machinery that exists in the parent codebase), and the dominant finite-size effects are the conventional ones of a near-critical or ordered magnet — tower-of-states structure and level discretization — for which the conventional tools are the right ones.

14.2 Beyond, in the XXZ model: the mask retires

Past the Heisenberg point in the XXZ model ($J_{\pm} < -0.5$), the transverse antiferromagnetic coupling dominates. There is *no* dual ice manifold to migrate to: the two transverse axes remain exactly degenerate ($J_x = J_y$), so no in-plane Ising arrows can be defined. The physics is that of a planar (XY) pyrochlore antiferromagnet — generically ordered, with Goldstone-like towers. Here the mask has nothing left to protect, and, pleasingly, *observable-level twist averaging finally works*: what needs converging there is ordinary level discretization, not a topological amplitude, and that is the problem twist averaging was always good at. Each tool retires exactly where the other’s jurisdiction begins.

14.3 Beyond, in the XYZ model: the mask reborn

The real cerium materials are not XXZ but XYZ: three independent couplings J_x, J_y, J_z . There the question “what happens when J_x or J_y dominates?” has a beautiful answer: the low-energy manifold is the two-in–two-out manifold *in the dominant axis’s basis*, and the XYZ model enjoys an exact **axis-permutation duality** — relabel the local axes and the J_x -dominant model becomes the J_z -dominant model. Everything transfers verbatim: the identity $\rho = \frac{1}{2} w \cdot L$ is lattice combinatorics that never asks which local basis defines the arrows; the sectors, the mask, the Feshbach fold, the BIC/Gamow taxonomy all rebuild in the rotated frame with zero new theory. The duality even donates a free exactness test: results computed on the J_x -dominant side must mirror the J_z -dominant side bit for bit.

What remains genuinely hard is the *crossover shell* around the symmetric point, where two (or three) descriptions coexist with comparable, low purities. In that no-man’s-land the

finite-size problem is not topological in *any* basis, and the axis-spread diagnostic above is the instrument that maps where the shell begins and ends.

The whole territory fits in one picture (Fig. 13): three handoffs, each at a boundary the campaign either measured or proved.

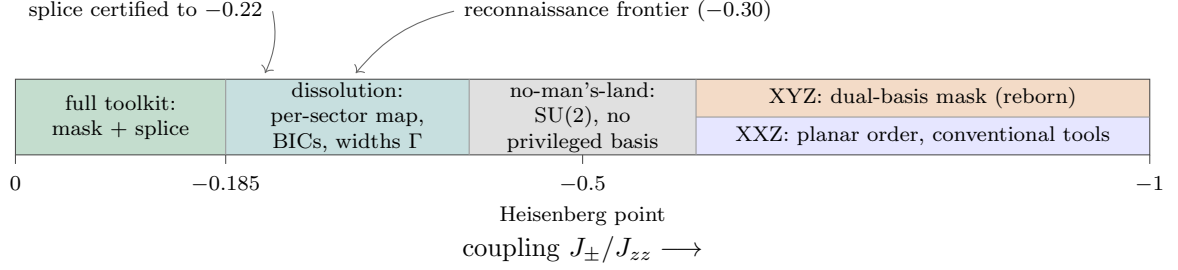


Figure 13: The regime map, with instrument handoffs. Green: weak coupling, where mask, counterterm, and splice all operate. Teal: the dissolution regime, where band physics belongs to the per-sector masked map (BICs and Gamow widths) and $H + C$ survives only as an estimate with quoted residue. Grey: the shell around the Heisenberg point, where SU(2) symmetry makes every ice basis a convention and no topological cleaning is meaningful. Beyond: in XYZ materials the whole machinery is reborn in the dominant axis's basis (orange); in strict XXZ the planar magnet takes over and conventional finite-size tools — including honest twist averaging — are the right ones (blue). Boundaries drawn at the 16-site values; their precise location is itself a finite-size quantity.

15 Wrong turns, kept on purpose

Science reporting usually deletes the dead ends. This campaign keeps them, because each one carries a transferable rule.

The wrong turn	The rule it taught
“Twist averaging removes winding contamination” (original claim, retracted)	Phases attach to virtual routes, not loop topology. Compute the object you claim vanishes.
The twist period was taken as 2π ; corrected to 4π	The transport quantum is <i>half</i> a box vector (Eq. 5); flux periods follow from the charge actually transported, not from habit.
“The label is emergent electric charge”	The ice manifold is charge-free; the label is the <i>magnetization displacement</i> . Same machinery, correct words — and the words matter when you predict what couples to it.
A spectral counting theorem was applied to the masked map, yielding 31 233 “levels” in a 12 870-state space	The theorem’s hypotheses (it applies to genuine sub-blocks of a Hermitian matrix) failed for the masked object. Check hypotheses <i>before</i> trusting a theorem’s output; a nonsense count was the only warning.
The “anchoring residue” peak (a week of theory built on it)	It was an unconverged solver (3 fixed-point rounds at convergence rate ~ 0.5). Converge to 10^{-12} before interpreting 10^{-2} .
Sector symmetrisation; catalogue counterterms	Corrections must be derived from the exact object, not from a plausible model of it; both “improvements” made things worse.
Sparse eigensolver used to count degenerate levels	It silently returned 4 of 7 copies of a degenerate level. Never use a lone sparse solve as a multiplicity referee.
Peak positions read off a 3000-point grid “agreed to 0.00%”	The grid step was 0.585%; the agreement was quantization, not accuracy. Refine and interpolate before quoting exponents.

16 Cast of characters

symbol / term	meaning
J_{zz}	the ice-rule interaction; the unit of energy; spinon cost scale
$J_{\pm}$	the quantum (flip) interaction; the one dial; negative for cerium pyrochlores
ice manifold	all spin patterns obeying two-in-two-out on every tetrahedron (90 states on 16 sites, 2970 on 32)
spinon	a tetrahedron violating the ice rule; the emergent electric charge; energy $\sim J_{zz}$
ring exchange	collective flip of a closed loop of spins that stays inside the ice manifold
$g_6 = 12 J_{\pm} ^3/J_{zz}^2$	strength of the hexagon ring exchange = the photon scale = the real physics
$g_4 = 4J_{\pm}^2/J_{zz}$	strength of the wrap-around four-site loop = the box artifact
winding w	how many times a loop wraps each glued direction; the complete topological label
transport ρ	magnetization displacement of a flip process; $\rho = \frac{1}{2}w \cdot L$ exactly
sector	family of ice states sharing one polarization value; 25 on 16 sites, 85 on 32
the mask	deletion of every amplitude connecting different sectors = exact removal of all winding processes
$F(z)$	the Feshbach map: the ice-block amplitude table with all spinon excursions folded in exactly, at asking-energy z
Z , ice weight	the fraction of an exact eigenstate living inside the ice manifold, $Z_n = \langle n P n \rangle = \ P n\rangle\ ^2 \in [0, 1]$ (also called the residue or purity); $1 - Z$ = spinon dressing = the error and convergence rate of everything approximate; sum rule $\sum_n Z_n = \text{Tr } P = 90$ over the whole spectrum
$H + C$	full-space Hamiltonian with the artifact counterterm; whole spectrum at once; residue grows as $1 - Z$
the splice	$H + C$ spectrum with its lowest 90 levels replaced by the exact masked roots; best of both
BIC	bound state in the continuum: a winding-free level above the continuum edge, forbidden to decay by transport symmetry
girth engineering	choosing the cell shape to maximize the shortest wrap-around loop, demoting the irreducible artifact
twist θ	phase on flips crossing the glued edge; useless for topology (Sec. 5), essential for densifying the spinon continuum (Sec. 13)
Gamow root	complex self-consistent energy $E - i\Gamma/2$, appearing when a level meets its <i>own</i> sector's continuum; Γ is a physical lifetime, not an error
mirror map $G(z)$	the Feshbach fold run the other way: ice space folded into the spinon block; cheap because PHP is diagonal
truncation level L	keep spinon states within L pair-flip moves of the ice manifold: the depth of virtual excursion followed; the L -ladder drift is the large-cluster error bar
bordered matrix M_s	sector ice block bordered by the (truncated) spinon space; Hermitian, so sector spectra come from sparse Lanczos with no anchor — the calibrated crossing instrument
two-sided fold	counterterm-free thermodynamics: remove the 90 highest- Z_n raw levels (wherever they sit), insert the masked roots at weight one; the mirror map supplies Z_n as the <i>identifier</i> — the weighted version fails (hijacked ensemble)
Z -rank gap	jump in ice weight between the last removed and first kept raw level; bimodal at weak coupling, closes by -0.20 — the continuous dial of band identity, and the obstruction theorem seen in data
third peak	the low- T $C(T)$ structure below the ring anomaly; three distinct causes — counterterm splitting (fake), internal band gap (real), ground-multiplicity change (real) — see Sec. 9.6
Heisenberg point	$J_{\pm} = -J_{zz}/2$: full $\text{SU}(2)$ symmetry; every ice basis becomes a convention and the mask becomes axis-dependent — the method's structural boundary
axis spread	disagreement between masked results built on the z, x, y bases; measures how much meaning the mask retains as the Heisenberg point approaches
axis-permutation duality	exact XYZ relabeling of local axes; transfers the entire machinery to

In plain words

The campaign in four sentences. A small glued simulation box invents a wrap-around quantum process that is stronger than the real photon physics of quantum spin ice, and the popular twist-averaging remedy provably cannot remove it. Measuring how far each process transports magnetization identifies the wrap-arounds exactly, and deleting them (the mask) is a provably maximal, provably safe cleaning that works at any coupling once combined with the Feshbach fold. With the artifact gone, the true photon scale, its flux law, its non-decay, and a previously hidden physical anomaly all emerge — and the remaining error is the box itself. The next campaign attacks that error on three independent fronts: the mask for topology, cell-shape optimization for the irreducible wrap-pairs, and twist averaging — reborn in its legitimate role — to give the virtual spinons a dense continuum to live in. The method's writ runs from weak coupling all the way to the Heisenberg point, where $SU(2)$ symmetry turns the description itself into a convention — and past it, in the XYZ models of the real materials, everything is reborn in the basis of the new dominant axis.